

BERNARDO CARVALHO

Software Developer

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Education

Centro Universitário Senac RS (UniSenac), Pelotas, RS, Brazil

Bachelor of Technology - BTech in System Analysis and Development | March 2024 – Present

Universidade Federal de Pelotas (UFPel), Pelotas, RS, Brazil

Bachelor's degree in Economics | 2017 – 2021

Colégio Gonzaga, Pelotas, RS, Brazil

High school diploma | 2014 - 2016

Experience

dti digital

Software Development Intern | December 2024 – Present

Active participation in the development of *LIA*, *Luggo's WhatsApp* chatbot (*Luggo* is a subsidiary of *MRV*, the largest civil construction company in Latin America), including development of new features (*C#*, *.NET Core*), preparation of unit tests (*NUnit*, *xUnit*) and integration with external systems (*Meta*, *IBM Watson*, *Oracle RightNow*), in addition to brief participation in the development of *dti evolve* (*C#*, *TypeScript* (*React*)), an internal monitoring squad performance platform.

Senatec — Centro Universitário Senac RS (UniSenac)'s junior company

Project Coordinator | December 2024 – Present

Coordinated projects such as *Fisiosul System*, which manages all hospital care for a physical therapy clinic, and the *Jaguarão Film Festival* website. Participated in in-person and remote meetings with real clients and conducted the *Senatec's* newest recruitment process.

Full-stack Developer | May 2024 – Present

Active participation in the development of several projects, notably the *Senatec System*, which aims to maintain records regarding the junior enterprise's employees and projects, and the already mentioned *Fisiosul System*, for managing appointments at a physiotherapy clinic, both with *TypeScript* (*React* and *Express.js*). Also contributing to the development of the pages of the *ARS Ecosystem* (*Audiovisual Southern Region*), carried out in partnership with the *Federal University of Pelotas (UFPel)* and the *Jaguarão Film Festival – RS*, both built in *WordPress*.

Languages

- English (fluent).
- Portuguese (native).
- Spanish (intermediate).

Featured project

Transaction API

<https://github.com/bgbaine/transaction-api>

Built with *Java* and *Spring Boot*, this *RESTful API* follows *Clean Architecture* principles, while also incorporating elements of *Hexagonal Architecture* and *Domain-Driven Design (DDD)*. It allows basic *CRUD* operations, such as creating and deleting transactions, calculating basic statistics, such as number of transactions, average transaction value and more. This project also contains a *Swagger* documentation, unit tests written with *Mockito*, metrics and *HealthCheck* provided by *Spring Actuator*, local containerization with *Docker* and a thorough documentation.

Hard skills

Programming

- Comfortable with *C/C++/C#, Java, JavaScript/TypeScript, Python, SQL*.
- Experience with unit testing frameworks such as *Jest, Mockito, Moq, NUnit, pytest, xUnit*.
- Experience with integration testing.

Web development

- Comfortable with *HTML, CSS, Markdown, WordPress*.
- Experience with frontend frameworks such as *Bootstrap, React, Tailwind CSS*.
- Experience with backend frameworks such as *ASP.NET Core, Express.js, Fastify, Flask, Spring Boot*.
- Experience with *ORMs* such as *Prisma, Sequelize*.
- Experience with *RDBMSs* such as *Microsoft SQL Server, MySQL, SQLite, PostgreSQL*.
- Experience with *NoSQL* databases such as *Azure Table Storage, MongoDB, Redis*.
- Experience with applications such as *Azure Storage Explorer, DBeaver, Node.js, Postman, SQL Server Management Studio*.
- Experience with *Continuous Integration and Continuous Delivery (CI/CD)* tools such as *Azure Pipelines*.

Version control

- Comfortable with *Git, GitHub, GitLab, Microsoft Azure Repos*.

Operating systems, virtualization and cloud computing services

- Comfortable with *Linux* distributions, *Windows*.
- Experience with *Docker, Kubernetes, VirtualBox, VMWare*.
- Experience with *AWS, Microsoft Azure* environments.
- Deploy and *VPS*' management experience with *Hostinger*.

Softwares and tools

- Comfortable with *Bash, DOS, Zsh* shells.
- Experience with code editors like *IntelliJ IDEA, Nano, Vim, Microsoft Visual Studio, VSCode*.
- Experience with project management tools like *Microsoft Azure DevOps, Trello*.
- Experience with information security tools like *Burpsuite, enum4linux, Netcat, Nmap, Metasploit, Wireshark*.
- Experience with office applications like *Google Docs, Google Sheets, Google Slides, Microsoft Excel, Microsoft PowerPoint, Microsoft Teams, Microsoft Word*.
- Experience with multimedia tools like *Adobe Photoshop, Figma, Mural, Vegas Pro*.

Courses

- CS50x - *CS50's Introduction to Computer Science* (2023).
- CS50P - *CS50's Introduction to Programming with Python* (2023).
- Santander's Bootcamp – *Back-end development with Java* (2024).
- Santander Coders – *Data Science with Python* (2024).
- TryHackMe – *Complete Beginner learning path* (2024).
- Amazon – *AWS Academy* (2025).
- Microsoft - Course AZ-900T00-A: *Introduction to Microsoft Azure* (2005).

Soft skills

- Communicative
- Fast learner
- Organized
- Proactive

Understanding of concepts

- Agile methodologies (Kanban, Scrum).
- Clean Code (*DRY, KISS, YAGNI*).
- Coding best practices.
- Cryptography (symmetric and asymmetric).
- Cybersecurity.
- Data structures (queues, stacks, linked-lists, trees, binary trees, hash tables and tries).
- Debugging.
- Frameworks.
- Hashing.
- Memory management (pointers, dynamic memory allocation, stack, heap and buffer overflow).
- Networks.
- Object-oriented programming (OOP).
- Procedural-oriented programming (POO).
- Programming logic.
- Protocols.
- Regular expressions.
- Requests, webhooks, service buses.
- *RESTful APIs*.
- Search algorithms (linear search and binary search).
- Software architectures (Clean Architecture, DDD, Hexagonal Architecture, Monolithic Architecture, MVC)
- Sort algorithms (bubble sort, selection sort and merge sort).
- Test Driven Development (TDD).
- Unit testing.

Other projects

CS50P's Final Project — Handball game in Python

<https://github.com/bgbaine/handball>

Handball clone, a game inspired by the famous *Atari's Pong*, and made for the *Coleco Telstar* (1976). Written in *Python*, makes use of the *Pygame* library. Uses concepts such as *classes*, *object-oriented programming (OOP)*, *unit tests (pytest)*, *type system (Mypy)*, *docstrings* formulation and formatting and *Markdown* language to elaborate a README file.

Codewars — Solved Katas

<https://github.com/bgbaine/codewars>

My solutions for multiple *Codewars' kata* (a.k.a. programming challenges) using multiple languages. Uses concepts such as *algorithms*, *coding best practices*, *programming logic*, *C*, *JavaScript* and *Python*.